



Forensic analysis of JPEG images

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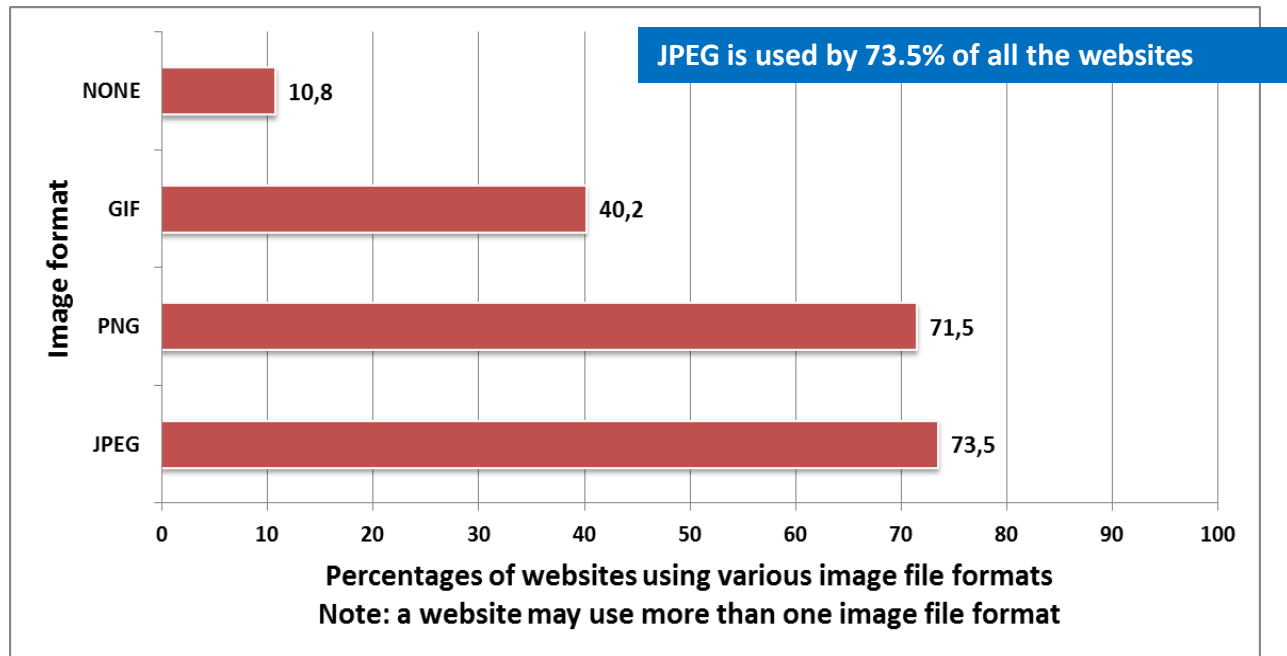
Agenda

- Introduction to JPEG compression
 - What is image compression?
 - The JPEG (Joint Photographic Expert Group) standard
- Forensic analysis of JPEG images
- Double JPEG image forensics
- Traces of image editing

Introduction to JPEG compression

What is JPEG?

- JPEG (Joint Photographic Expert Group) is an international standard for **lossy image compression** released in 1992
- JPEG is still today one of the most popular image formats on the Web



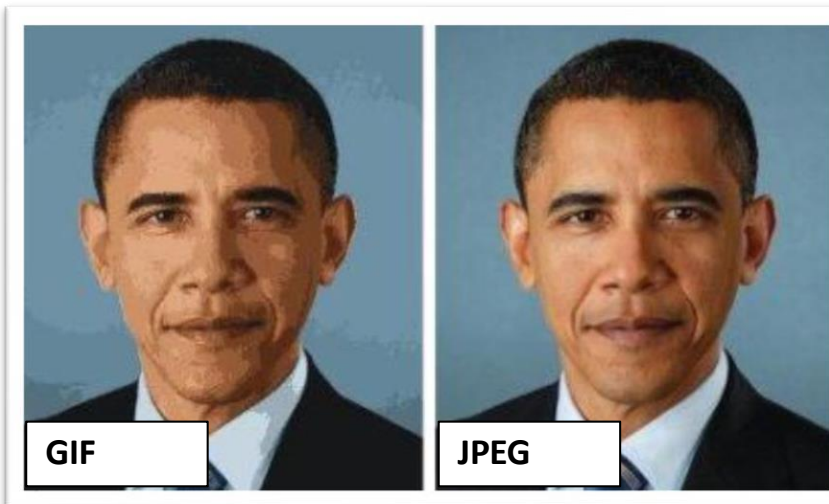
Source:
<https://w3techs.com/technologies/overview/image-format/all> (updated April 2016)

- Photos in social networks are in (lossy) compressed formats. Most of them are in JPEG format

Introduction to JPEG compression

What is JPEG?

- JPEG is used in many applications. It is particularly suitable for the compression of **photos and paintings of realistic scenes** with smooth variations of tone and color



- With respect to the other widely diffused GIF format JPEG ensures **better visual quality** compressed images **for the same file size**

Introduction to JPEG compression

Importance of compression (in real life)

File type	Size
Tiff, uncompressed	901K
Tiff, LZW lossless compression (yes, its actually bigger)	928K
JPG, High quality	319K
JPG, medium quality	188K
JPG, typical web quality	105K
JPG, low quality / high compression	50K
JPG, absurdly high compression	18K
PNG, lossless compression	741K
GIF, lossless compression, but only 256 colors	286K

Introduction to JPEG compression

Impact of compression (in real life)



Higher compression at the price
of a lower visual quality

Introduction to JPEG compression

Why/How can images be compressed?

- Image compression can be achieved because image data are **often highly redundant and/or irrelevant**.
- Image coding is achieved by *reducing the redundancy* contained in data. More specifically, two kinds of redundancy exist:
 - *statistical redundancy*, which is exploited for **lossless compression**
 - *Irrelevance (psychovisual redundancy)*, whose removal leads to **lossy compression**

Introduction to JPEG compression

Statistical redundancy

- Spatial redundancy
 - correlation between neighboring pixels
- Spectral redundancy
 - correlation among color components
- Temporal redundancy (for video compression)
 - correlation between consecutive frames

Introduction to JPEG compression

Spatial redundancy: an example

- The difference between two adjacent pixels has a very skewed distribution centered around 0



Introduction to JPEG compression

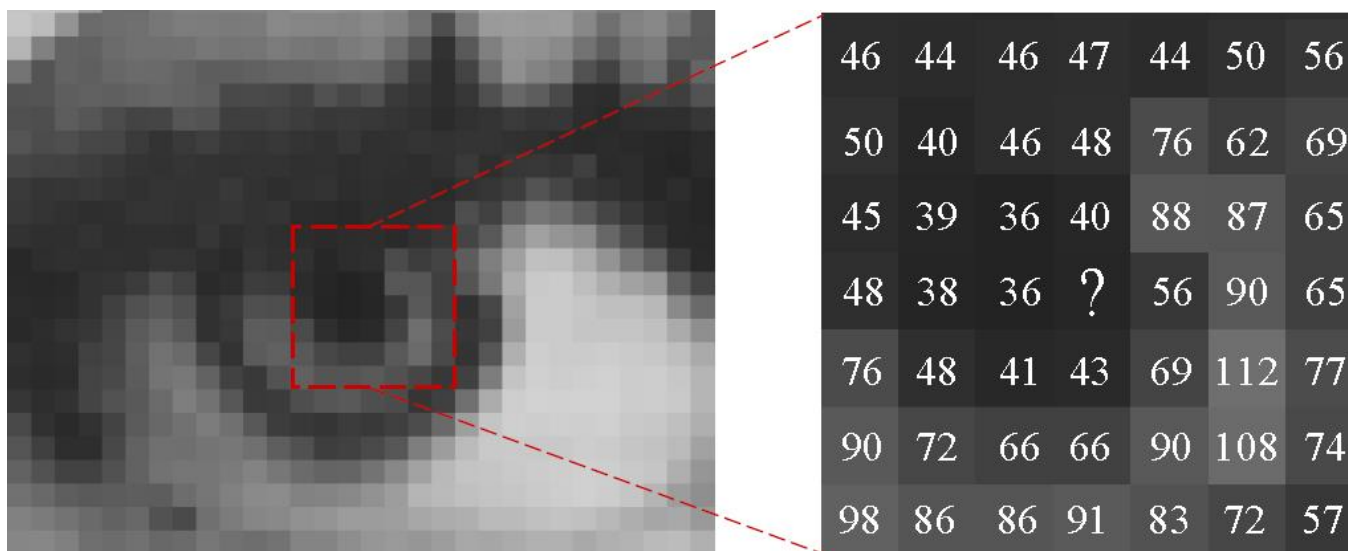
Psychovisual redundancy

- Spatial irrelevance
 - refers to the ability of the Human Visual System (HVS) to perceive small image details
- Spectral irrelevance
 - refers to the way the HVS perceives colors
- Temporal irrelevance (for video compression)
 - accounts for the ability of the HVS to perceive rapid changes between subsequent video frames

Introduction to JPEG compression

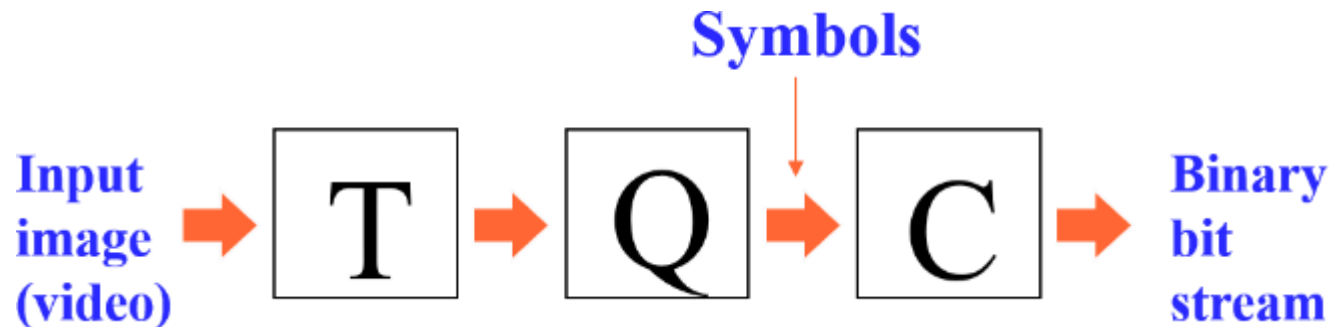
Spatial redundancy and irrelevancy

- What is the value of the missing pixel? **(39)**
- How critical is the exact reproduction?



Introduction to JPEG compression

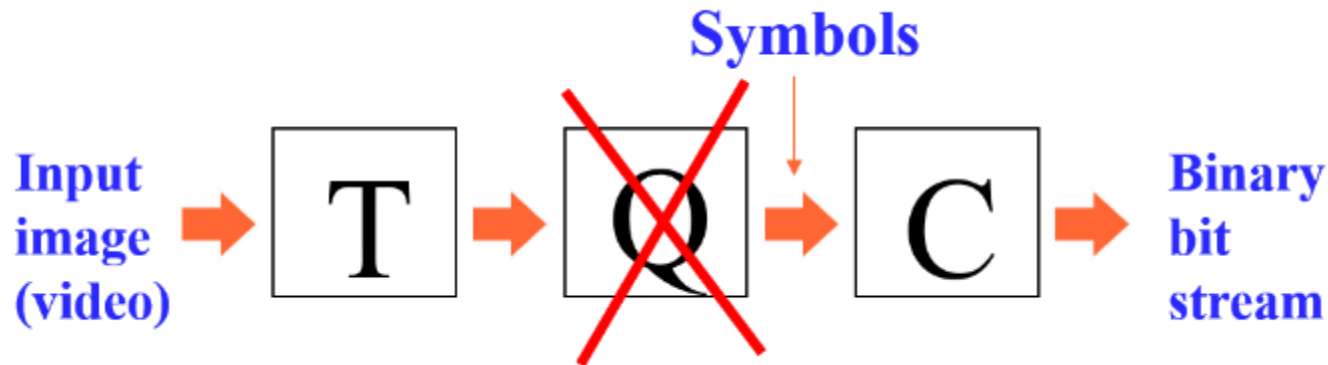
General compression scheme



- **T = Transformer**, it applies a one-to-one transform to input data, the output should be more amenable to compression (e.g. skew probability distribution, reduced correlation among data). No loss occurs here. Examples: predictive mapping, DCT transform.
- **Q = Quantizer**, it achieves lossy compression by performing a many-to-one mapping of data into symbols (scalar or vector quantization)
- **C = Coder**, by assigning a codeword to each symbol produced by the quantizer, lossless compression is achieved (fixed-length or variable-length codes may be used)

Introduction to JPEG compression

Lossless compression

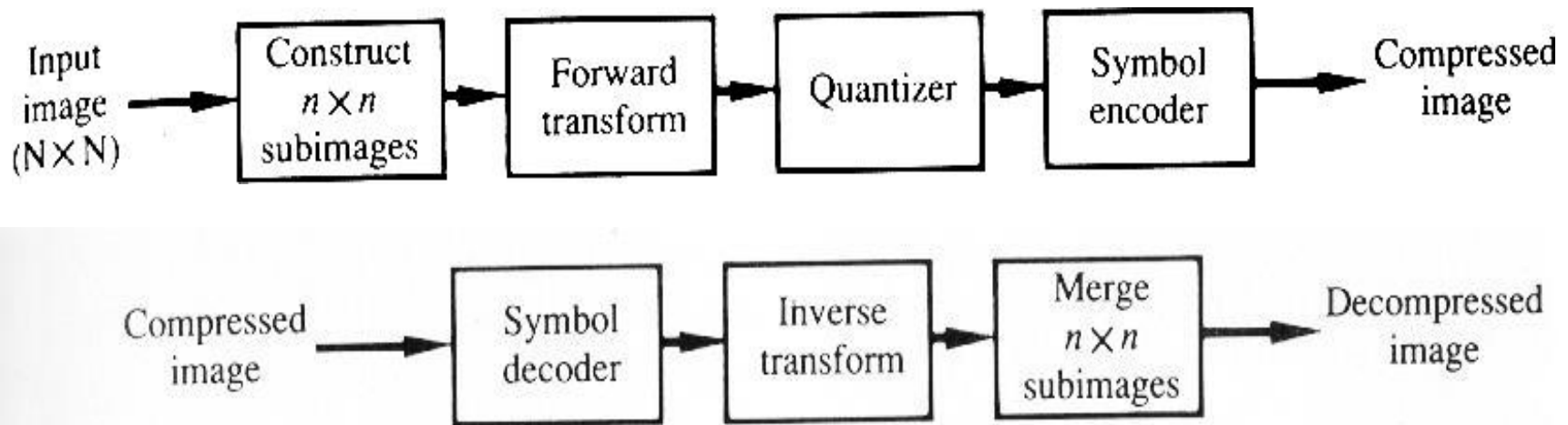


- By removing the quantizer, a general lossless compression scheme is obtained
- The transformer **T** only aims at removing the spatial, spectral and temporal redundancy (memory), or at putting it in a different form, so that it is easier for the symbol coder to compress data
- Compression ratios achievable through lossless coding are not sufficient to meet the needs of most practical applications

Introduction to JPEG compression

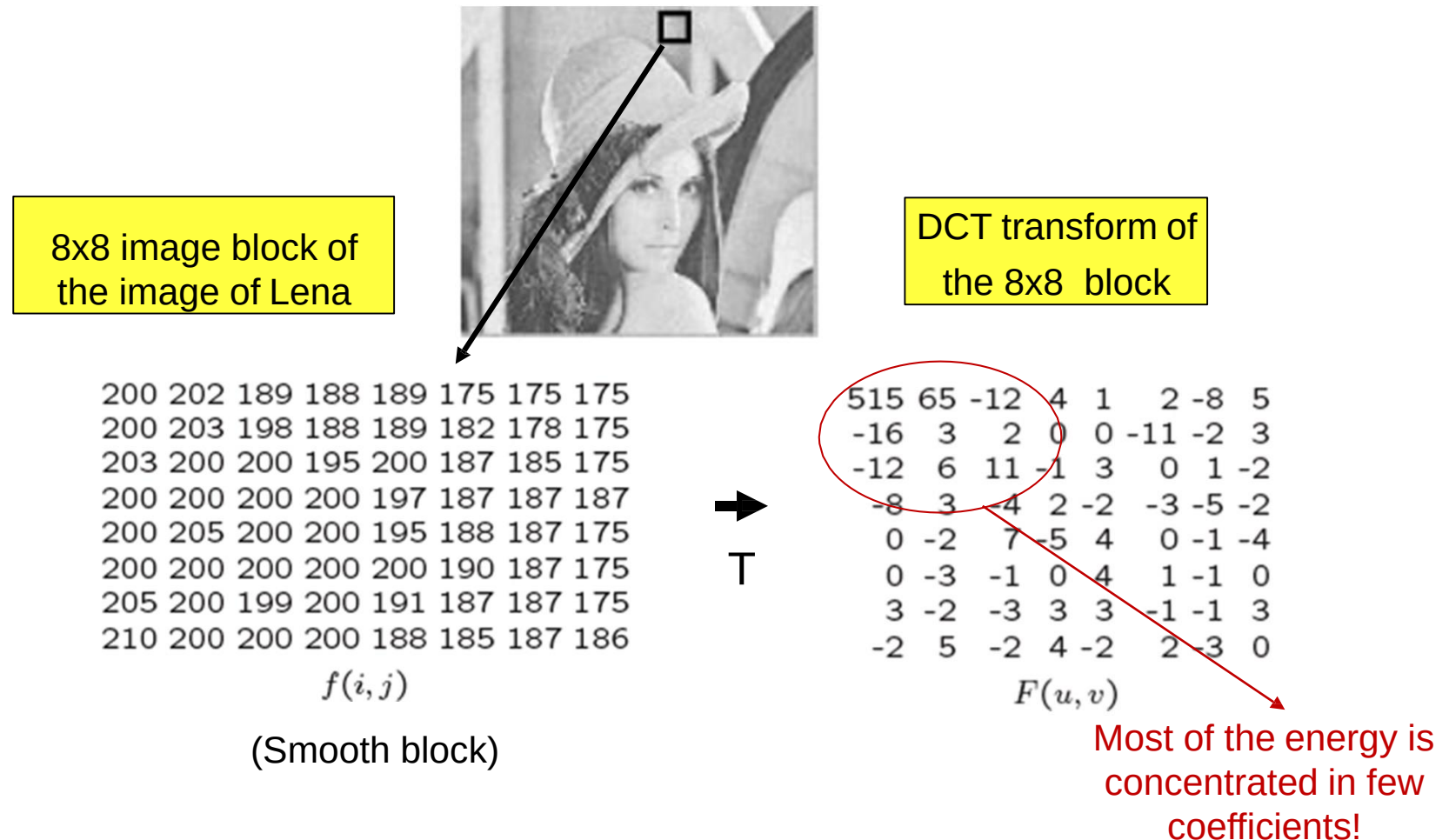
Block-based transform (T)

- Transform coding is performed by taking an image and breaking it down into sub-image (block) of size $n \times n$. The transform is then applied to each sub-image (block) and the resulting transform coefficients are quantized and entropy coded.



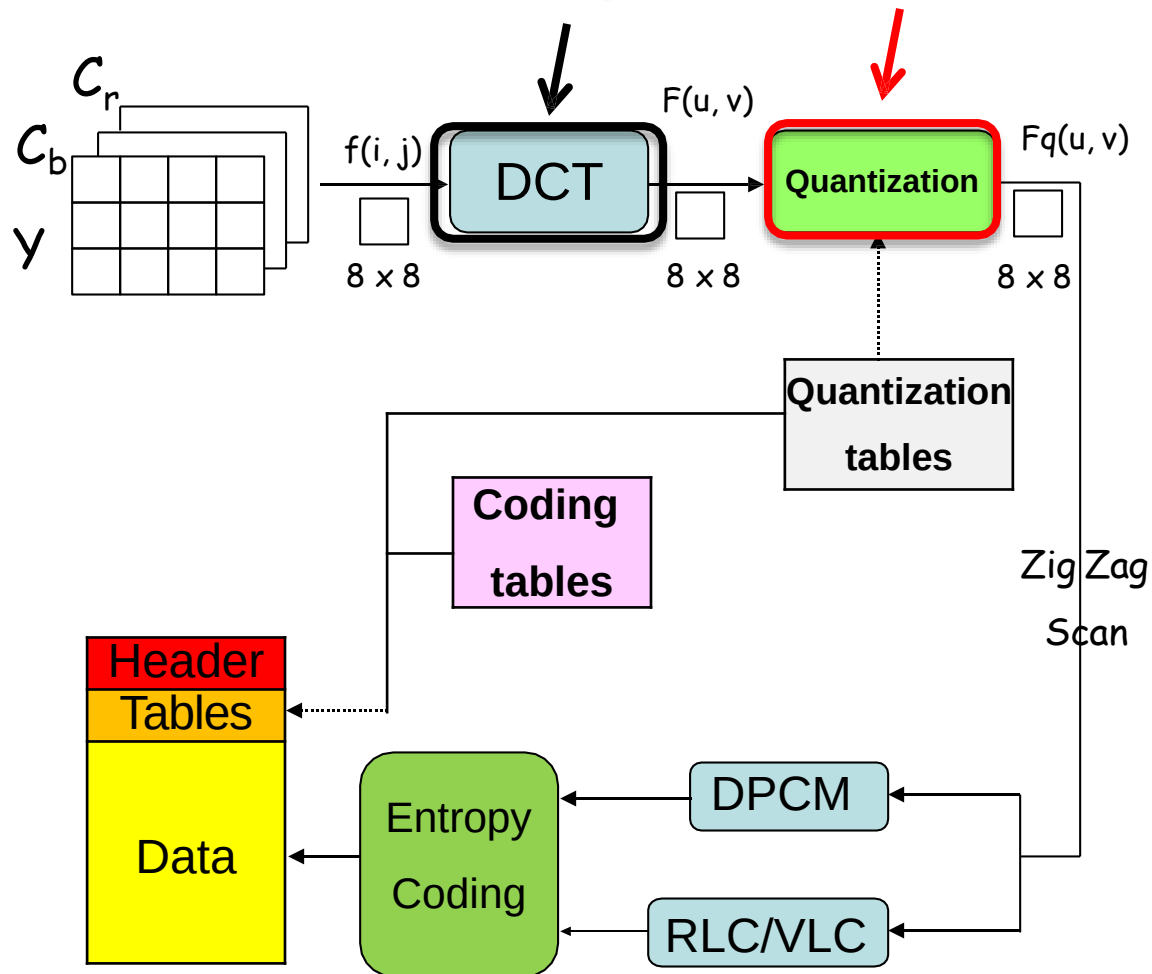
Introduction to JPEG compression

An example



Introduction to JPEG compression

JPEG baseline encoding



Main steps:

1. Discrete Cosine Transform (DCT) of each 8x8 pixel block
2. Scalar quantization
3. Zig-zag scan to exploit redundancy
4. Data preparation for entropy coding:
 - differential pulse-code modulation (DPCM)
 - run-length coding (RLC)
 - variable-length coding (VLC)
5. Entropy coding

Reverse order for decoding

Introduction to JPEG compression

Color space transform: RGB to YCbCr

- **RGB** color space is not the only method of image representation
- There are several other color spaces, each one with its properties
- A popular color space in image compression is the **YCbCr**, which:
 - separates *luminance* (Y) from *color information* (Cb,Cr)
 - processes Y and (Cb,Cr) separately (not possible in RGB)
- RGB to YCbCr (and YCbCr to RGB) linear conversions:

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \quad \begin{array}{l} Y \in [0, 255] \\ C_b \in [0, 255] \\ C_r \in [0, 255] \end{array}$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1.000 & 0.000 & 1.403 \\ 1.000 & -0.344 & -0.714 \\ 1.000 & 1.773 & 0.000 \end{bmatrix} \cdot \begin{bmatrix} Y \\ C_b - 128 \\ C_r - 128 \end{bmatrix} \quad \begin{array}{l} R \in [0, 255] \\ G \in [0, 255] \\ B \in [0, 255] \end{array}$$

Introduction to JPEG compression

Color space transform – example



Original



Red



Green



Blue



Original



Luma (Y)



Chroma (C_B)



Chroma (C_R)

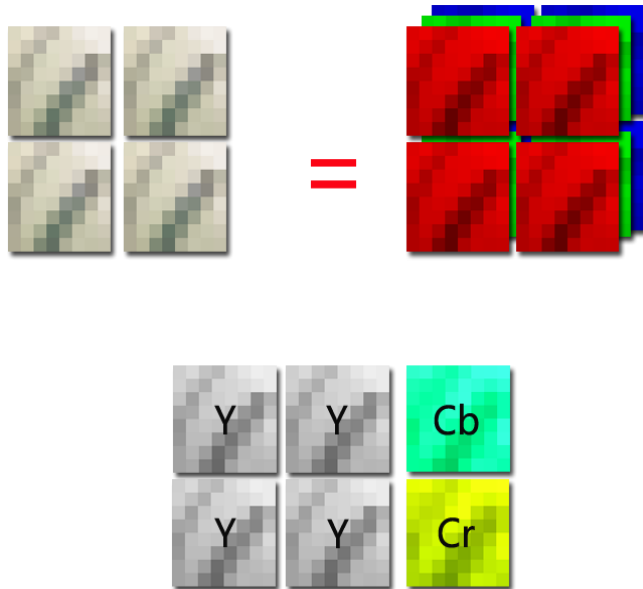
Introduction to JPEG compression

Color space transform – subsampling

- Y is taken every pixel, and Cb,Cr are taken for a block of 2x2 pixels

Data size is reduced to a half without significant losses in visual quality

- Example: block 64x64



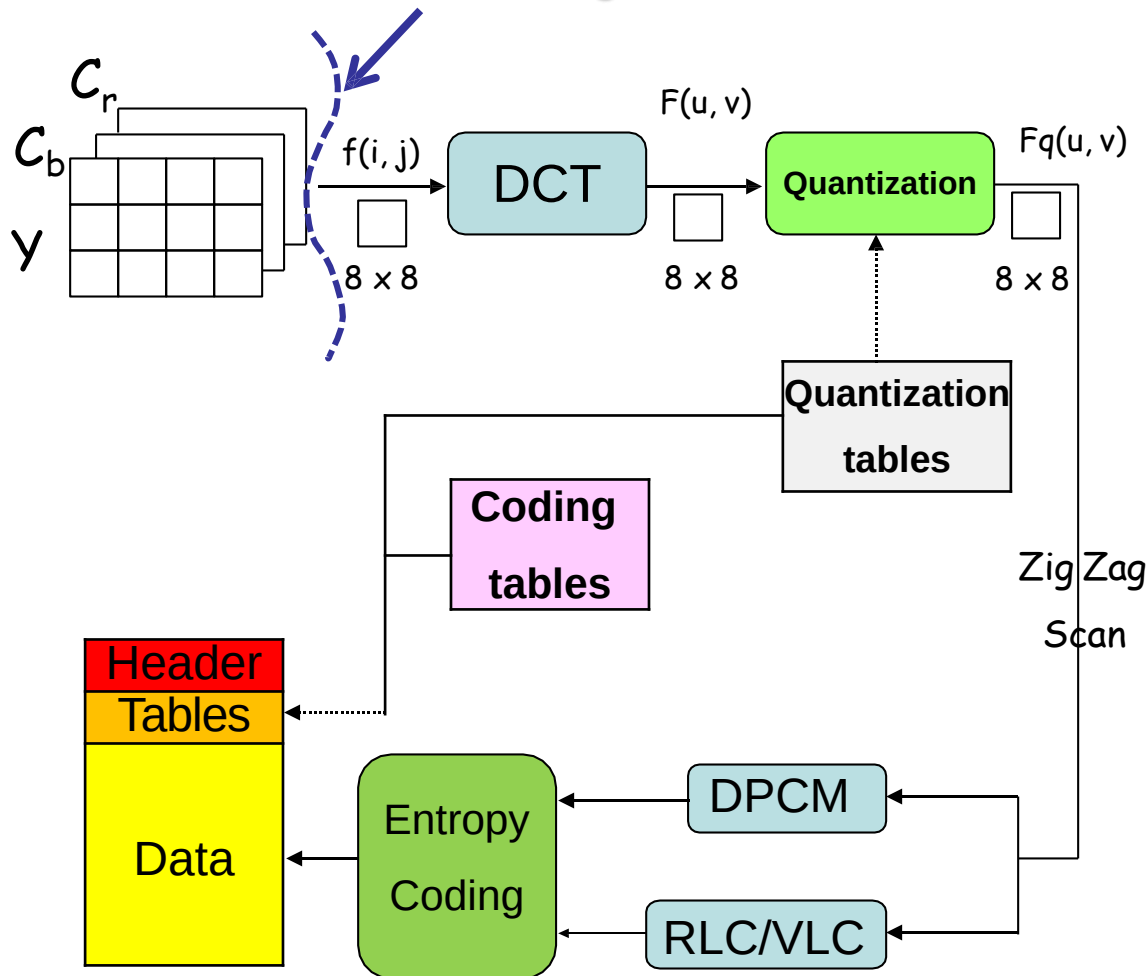
Without subsampling, one must take 64^2 pixel values for each color channel:
 $3 * 64^2 = 12288$ values (1 bytes per value)

JPEG takes 64^2 pixel values for Y channel and $2 * 32^2$ pixel values for chroma channel:

$64^2 + 2 * 32^2 = 6144$ values (1 bytes per value)

Introduction to JPEG compression

JPEG baseline encoding



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Reverse order for decoding

Introduction to JPEG compression

Discrete Cosine Transform (DCT)

- **Transformed data are more suitable to compression** (e.g., skew probability distribution, reduced correlation).
- 2D-DCT is used for image (2D-signal) compression

Forward DCT

$$F(u, v) = \frac{1}{4} C(u) C(v) \sum_{x=0}^7 \sum_{y=0}^7 f(x, y) \cos \left[\frac{\pi(2x+1)u}{16} \right] \cos \left[\frac{\pi(2y+1)v}{16} \right]$$

for $u = 0, \dots, 7$ and $v = 0, \dots, 7$

where $C(k) = \begin{matrix} 1/\sqrt{2} & \text{for } k=0 \\ 1 & \text{otherwise} \end{matrix}$

Inverse DCT

$$f(x, y) = \frac{1}{4} \sum_{u=0}^7 \sum_{v=0}^7 C(u) C(v) F(u, v) \cos \left[\frac{\pi(2x+1)u}{16} \right] \cos \left[\frac{\pi(2y+1)v}{16} \right]$$

for $x = 0, \dots, 7$ and $y = 0, \dots, 7$

Introduction to JPEG compression

2D-DCT: computation

Pixel block

214	224	199	190	222	231	239	236
215	198	179	218	219	223	229	236
202	166	205	215	218	208	144	183
173	177	211	212	200	152	136	196
155	197	205	200	143	133	178	221
158	197	198	153	118	159	202	211
169	192	167	110	139	190	196	199
169	180	131	105	172	193	190	191

- 128 =

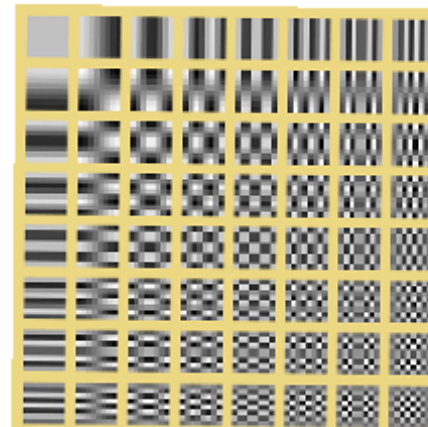
86	96	71	62	94	103	111	108
87	70	51	90	91	95	101	108
74	38	77	87	90	80	16	55
45	49	83	84	72	24	8	68
27	69	77	72	15	5	50	93
30	69	70	25	-10	31	74	83
41	64	39	-18	11	62	68	71
41	52	3	-23	44	65	62	63

Shift operations:
[0, 255] -> [-128, 127]

$$S_{vu} = \frac{1}{4} C_u C_v \sum_{x=0}^7 \sum_{y=0}^7 s_{yx} \cos \frac{(2x+1)u\pi}{16} \cos \frac{(2y+1)v\pi}{16}$$

DCT
Result

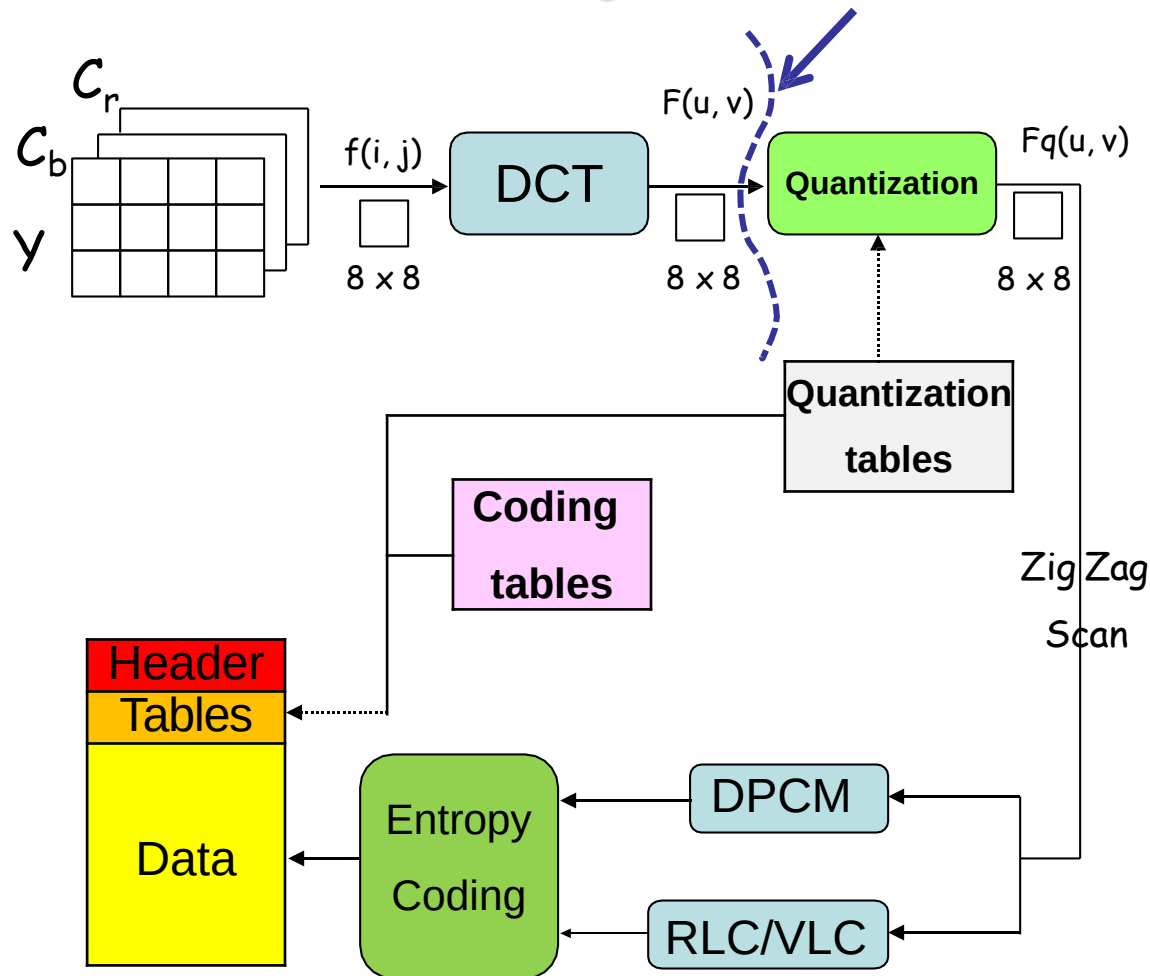
479	-35	52	-30	-7	-27	10	-3
141	11	-62	11	56	14	5	-5
41	-54	48	98	-35	-19	-14	8
22	-16	56	-48	-48	17	-24	-3
-4	5	-6	-16	14	-15	-15	9
-10	20	-11	9	-7	-15	22	12
-9	12	-12	5	-4	0	14	3
-2	6	-2	3	-3	0	8	-2



Meaning of
each position
in DCT result-
matrix

Introduction to JPEG compression

JPEG baseline encoding



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Reverse order for decoding

Introduction to JPEG compression

Quantization

- **Goal: to reduce number of bits per sample**
- For each 8x8 DCT block, $F(u,v)$ is divided by a 8x8 **quantization matrix** Q

$$F_q(u, v) = \text{round} \left(\frac{F(u, v)}{Q(u, v)} \right)$$

$$\hat{F}(u, v) = F_q(u, v) \cdot Q(u, v)$$

(Reconstructed value)

$$Err(u, v) = \hat{F}(u, v) - F(u, v)$$

(Reconstruction error)

$Q(u,v)$, quantization step
at frequency (u,v)

- Example (one number): $F = 45$
 - $Q = 4$: $F_q = \text{round}(11.25) = 11$ (De-quantize: $11 \times 4 = 44$, against 45. **Err = 1**)
 - $Q = 8$: $F_q = \text{round}(5.625) = 6$ (De-quantize: $6 \times 8 = 48$, against 45. **Err = 3**)
- **Quantization error is the main reason why JPEG compression is LOSSY**

Introduction to JPEG compression

Quantization

- Each $F[u,v]$ in a 8x8 block is divided by constant value $Q(u,v)$
- **Higher values in the quantization matrix Q allows to achieve *better compression* at the cost of *visual quality***
- How to choose Q ?
- Eye is *more sensitive to low frequencies* (upper left corner of the 8x8 matrix), *less sensitive to high frequencies* (lower right corner)
- **Idea: quantize more (large quantization step) the high frequencies, less the low frequencies**
- The values of the Q matrix are controlled with parameter called **Quality Factor (QF)**.
 - QF ranges from 100 (best quality) to 1 (extremely low)

Introduction to JPEG compression

Quantization table: luminance

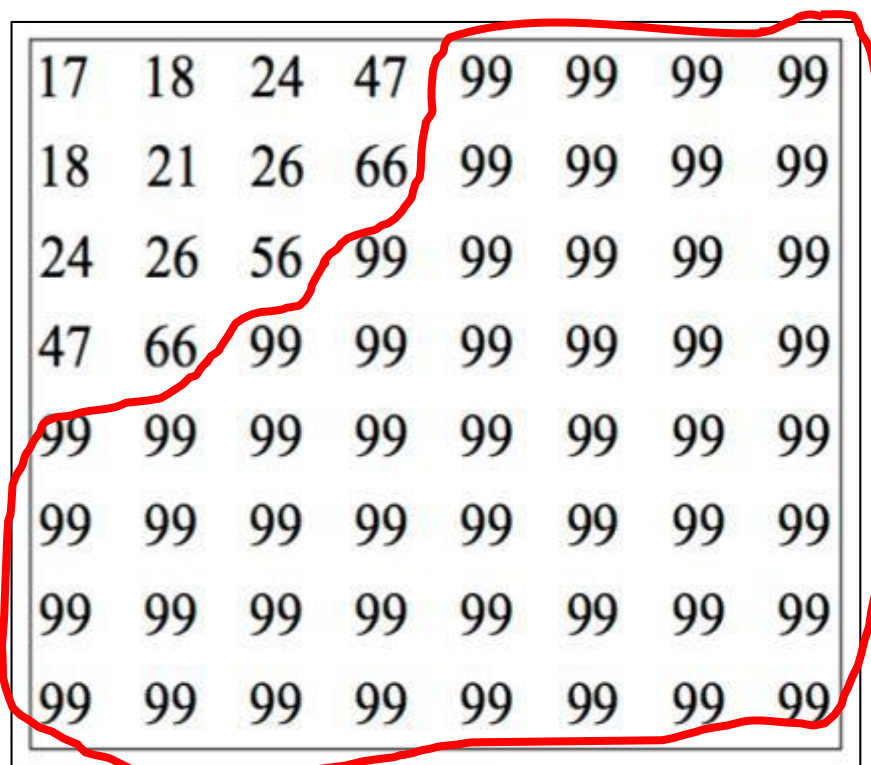
- Example: Quantization table Q for **QF = 50**

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	36	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

Introduction to JPEG compression

Quantization table: chrominance

- Example: Quantization table Q for **QF = 50**
- Colors can be quantized more coarsely due to reduced sensitivity of the Human Visual System (HVS)



17	18	24	47	99	99	99	99
18	21	26	66	99	99	99	99
24	26	56	99	99	99	99	99
47	66	99	99	99	99	99	99
99	99	99	99	99	99	99	99
99	99	99	99	99	99	99	99
99	99	99	99	99	99	99	99
99	99	99	99	99	99	99	99

Introduction to JPEG compression

Quantization: luminance and chrominance

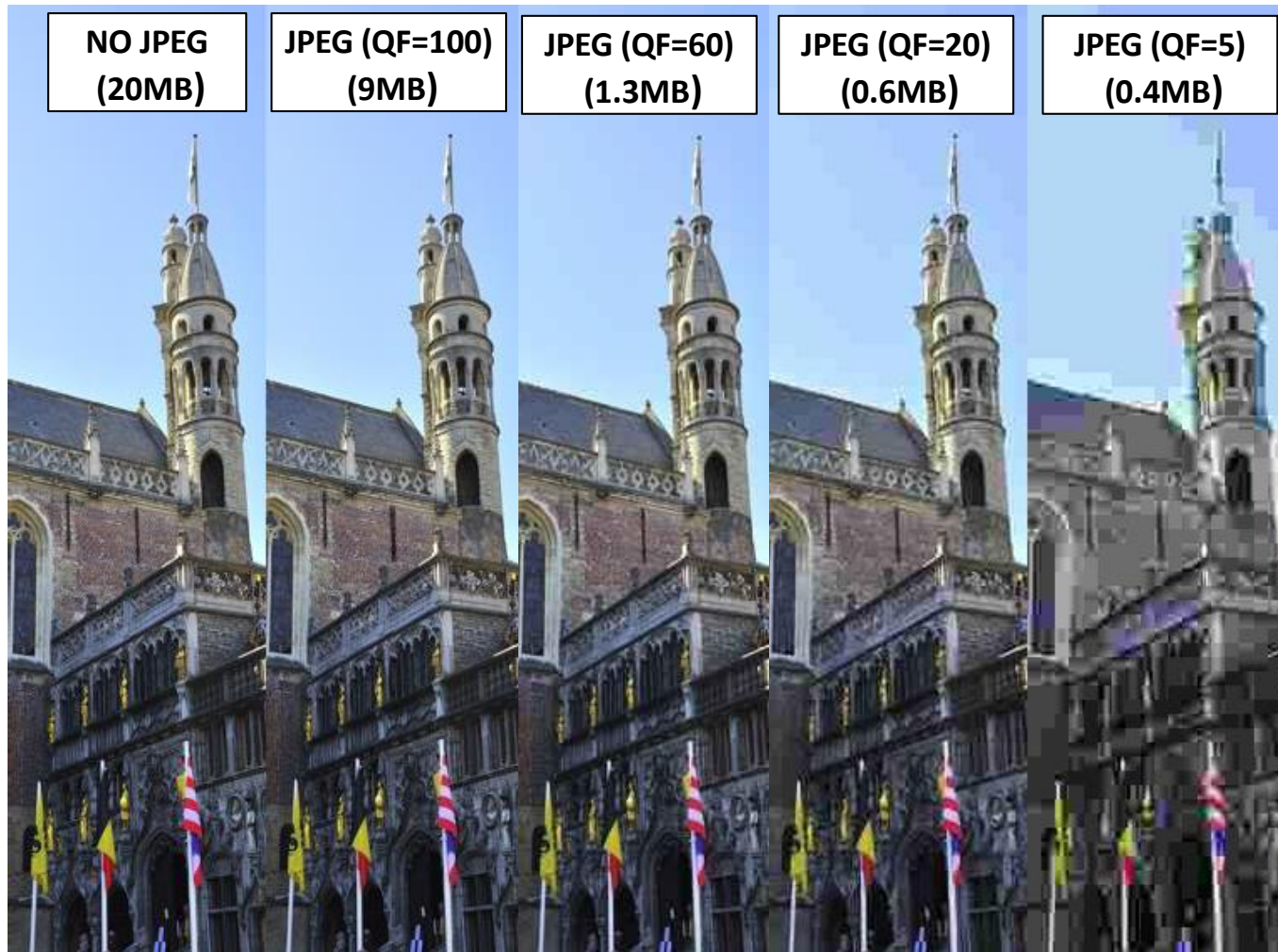
- An example of quantization table Q for **QF = 70**

Quantization Table: Luminance							
10	7	6	10	14	24	31	37
7	7	8	11	16	35	36	33
8	8	10	14	24	34	41	34
8	10	13	17	31	52	48	37
11	13	22	34	41	65	62	46
14	21	33	38	49	62	68	55
29	38	47	52	62	73	72	61
43	55	57	59	67	60	62	59

Quantization Table: Chrominance							
10	11	14	28	59	59	59	59
11	13	16	40	59	59	59	59
14	16	34	59	59	59	59	59
28	40	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59

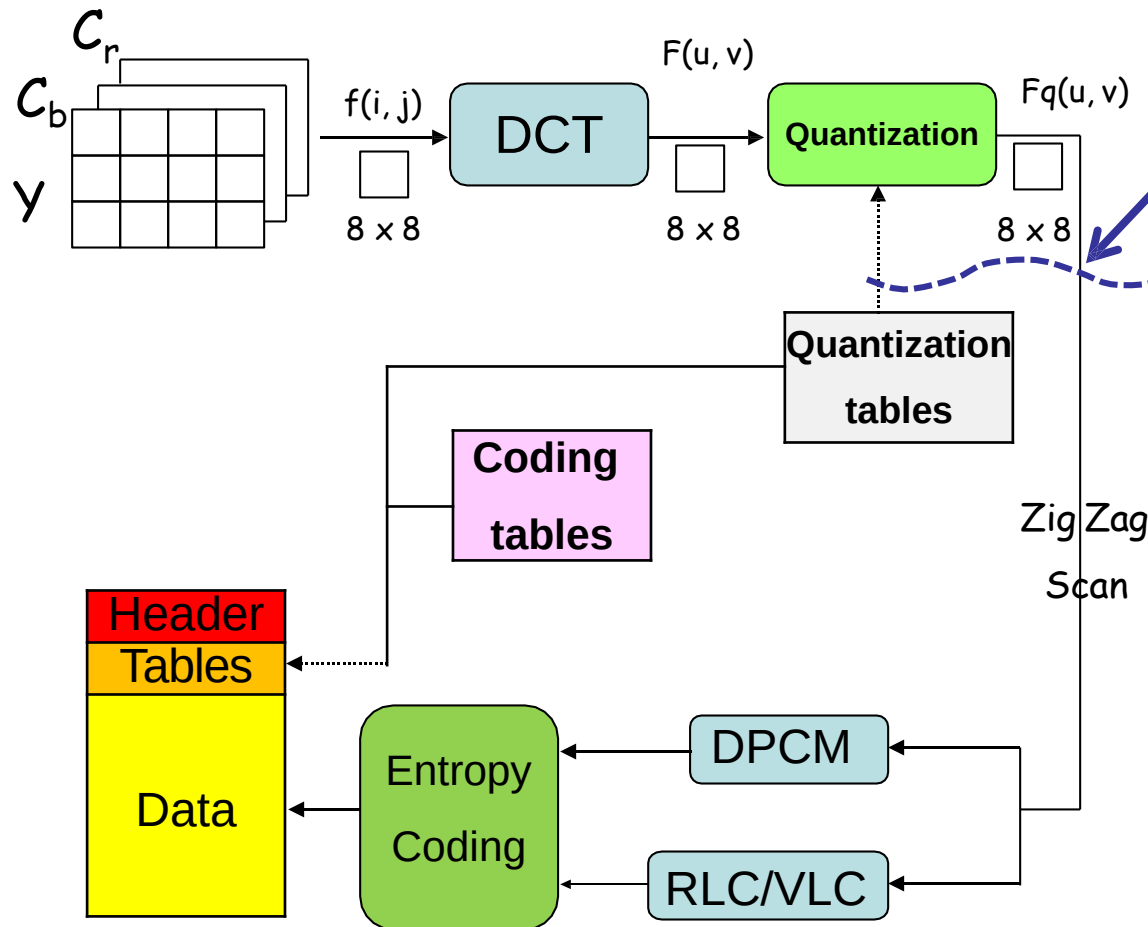
The quantization is less strong at larger QF

Introduction to JPEG compression



Introduction to JPEG compression

JPEG baseline encoding



Main steps:

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Reverse order for decoding

Introduction to JPEG compression

Preparation for Entropy Coding

- We have seen two main steps in JPEG coding: DCT transform (T) and quantization (Q)
- The remaining steps all lead up to entropy coding (C) of the *quantized block-DCT coefficients*:
 - ◆ these additional data compression steps are *lossless*
 - ◆ all the lossiness is in the quantization step

Introduction to JPEG compression

Remarks on JPEG compression

JPEG is effective because of the following main points:

- *Image data usually changes slowly across an image, especially within an 8x8 block*
 - therefore, images contain *much redundancy*
- *Experiments indicate that humans are not very sensitive to the high frequency data images*
 - therefore, we can remove much of this data exploiting transform coding
- *Humans are much more sensitive to brightness (luminance) information than to color (chrominance)*
 - JPEG performs subsampling of chrominance information (color channels)

Forensic analysis of JPEG images

JPEG compression footprints

- Like any other image processing, **JPEG leaves traces into the image**, especially at low Quality Factors
 - Such traces can be exploited to **gather useful information on the image**
- Some JPEG artifacts are immediately identified
 - **Blocking** due to block discontinuities
 - **Ring**ing on edges due to the DCT
 - **Graininess** due to coarse quantization
 - **Blurring** due to high frequency removal
- **Other (statistical) alterations are more subtle to identify!**

Forensic analysis of JPEG images

Blocking artifacts

- Processing each 8x8 block independently introduces **discontinuities along the block boundaries**, thus making image tiling visible

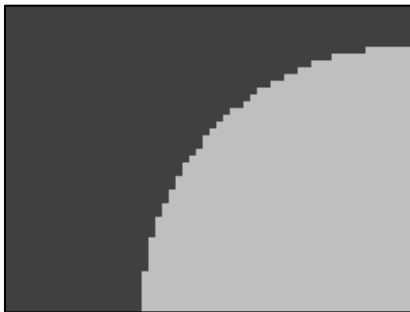


Forensic analysis of JPEG images

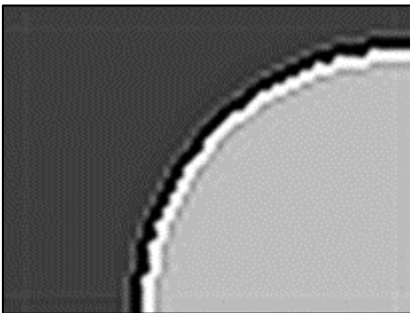
Ringing artifacts

- Spurious signals **near sharp transitions**
 - Visually, they appear as bands or “ghosts”
 - Particularly evident along edges as in text images

No ringing



Ringing



- A simple PDF file has four sections to it:
 - A one-line **Header** that identifies the file as a PDF file.
 - A **Body** that contains the document contents.
 - A **Cross-reference Table** that specifies the location of every indirect object in the file.

Forensic analysis of JPEG images

Graininess artifacts

- Particularly evident as “dots” along the edges



Forensic analysis of JPEG images

Blurring artifacts

- Removing high frequency DCT coefficients increases the smoothness of the image and is retaining shapes but making textures less distinguishable
 - **Human eye is particularly good at spotting smoothness**



Double JPEG image forensics

Double JPEG compression forensics

- Double JPEG compression is when an image is JPEG compressed first with QF_1 and then JPEG compressed again with QF_2
- In multimedia forensics, several approaches have been proposed to reveal **the footprints left by double compression**

Why knowledge that image has been JPEG compressed (quantized) twice is important?

Double JPEG image forensics

Suppose this landscape picture was taken by camera and image did not undergo any compression (TIF image, for example)



Double JPEG image forensics

Download an image from the Internet. It is very likely that it will be JPEG image compressed with a certain QF



With image editing software



Double JPEG image forensics

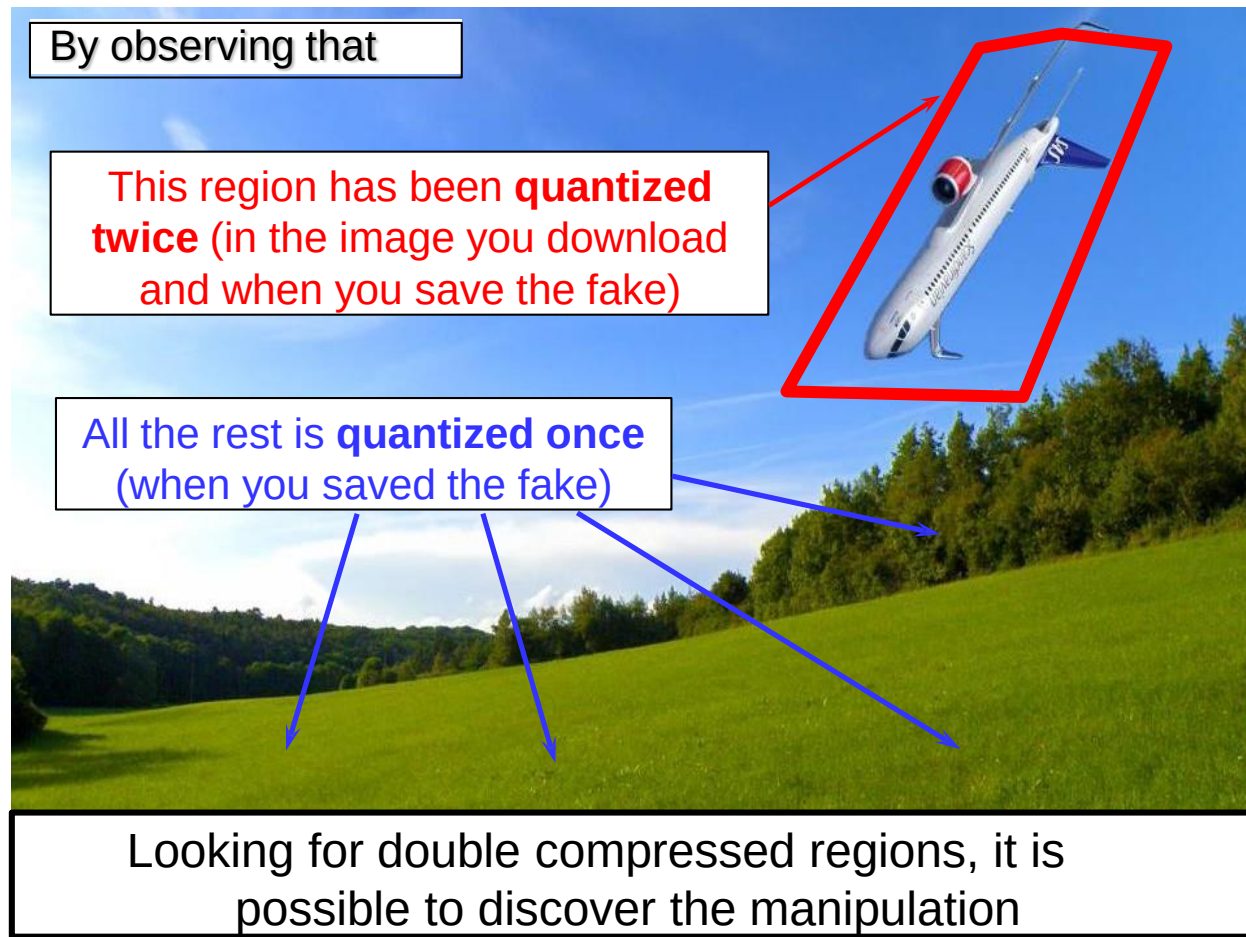
Created fake is realistic and deceptive image that usually will be stored **as JPEG**



How can one reveal your manipulation?

Double JPEG image forensics

Created fake is realistic and deceptive image that usually will be stored **as JPEG**



Double JPEG image forensics

Double JPEG compression: footprints

Why knowledge that image has been JPEG compressed (quantized) twice is important?

Double compression is an indicator of possible image manipulations

Double JPEG image forensics

Double quantization: footprints

- When an image is JPEG compressed first with QF_1 and then JPEG compressed again with QF_2 , a **double quantization** occurs
- Statistical footprints are left by double quantization
- Therefore, double JPEG images show these artifacts, while single JPEG does not
- Double JPEG detection can be performed based on these artifacts[*]

[*] Popescu, Alin C., and Hany Farid. "Statistical tools for digital forensics. Information *Hiding*. Springer Berlin Heidelberg, 2004.

Double JPEG image forensics

Single quantization (SQ)

- *Quantization* is the point-wise operation:

$$Q_a(x) = \text{round} \left(\frac{x}{a} \right)$$

where:

- a is a strictly positive integer (quantization step)
 - the value x/a is approximated to the closest integer
-
- *De-quantization* brings the quantized values back to their original range

$$\hat{x}_a = Q_a(x) \cdot a$$

Qa is not invertible because of the rounding operation

Double JPEG image forensics

Double quantization (DQ)

- Double quantization is a point-wise operation:

$$Q_{ab}(x) = \text{round} \left(\text{round} \left(\frac{x}{b} \right) \cdot \frac{b}{a} \right)$$

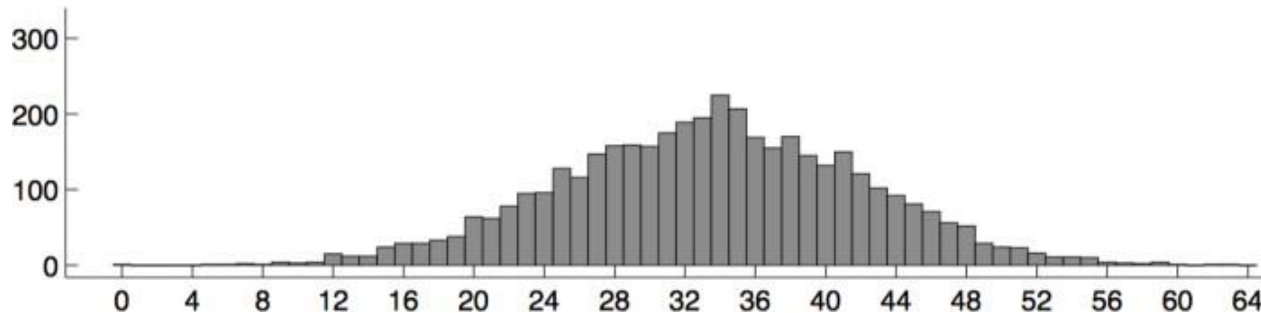
where b and a are the quantization steps of the first and second quantization

- Double quantization can be represented as a sequence of three steps:
 1. Quantization with step b
 2. De-quantization with step b
 3. Quantization with step a

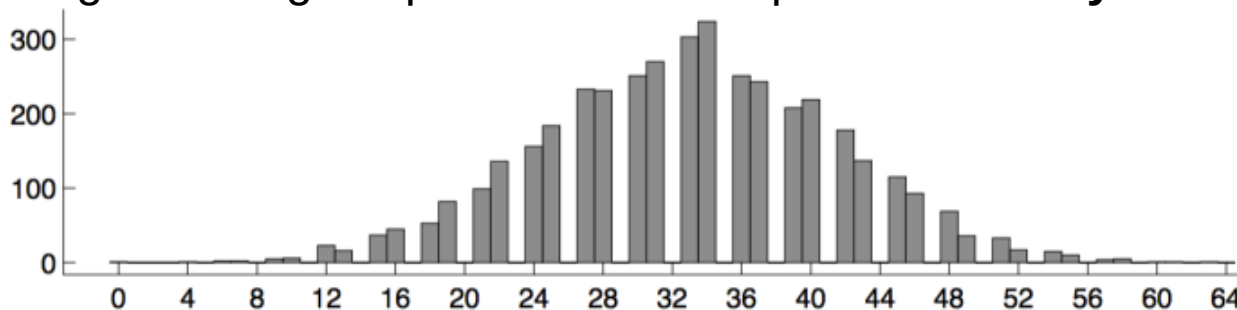
Double JPEG image forensics

Double quantization footprints (1/2)

- Consider a signal x with samples normally distributed in $[0,127]$.
- The histogram of the signal quantized with step **2** is the following:



- The histogram of signal quantized with step **3 followed by 2** is:

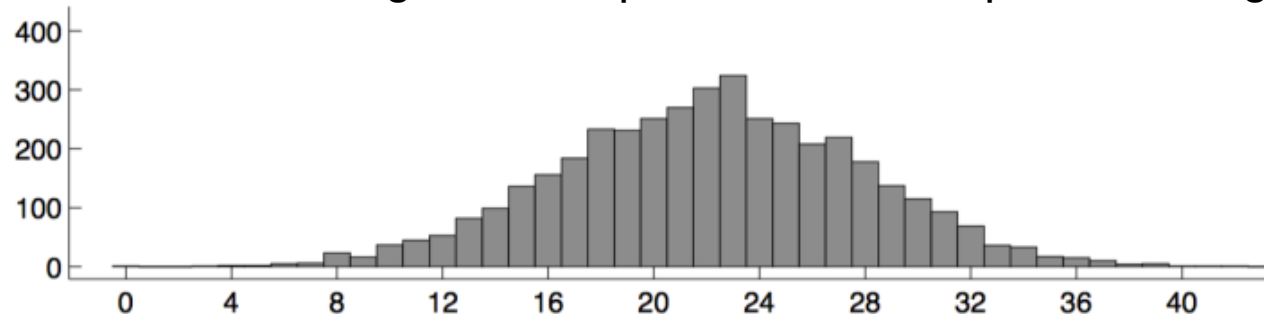


When $a < b$, some bins are empty (holes). This happens because the second quantization re-distributes the quantized coefficients into more bins than the first quantization

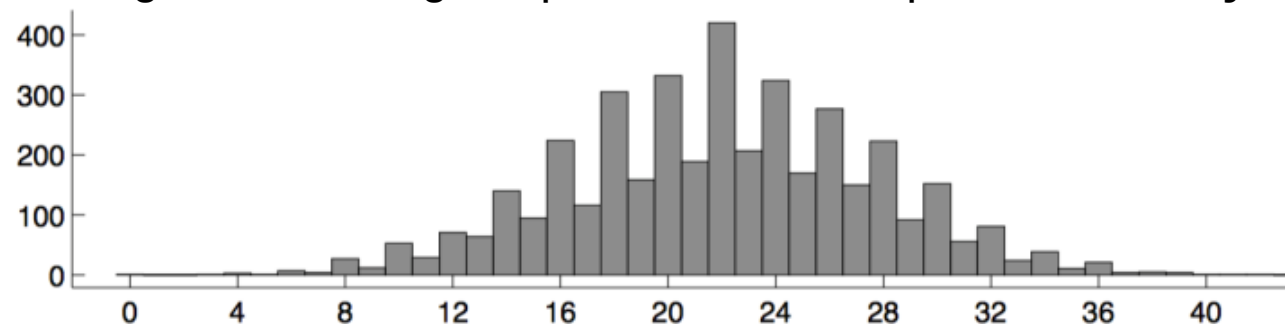
Double JPEG image forensics

Double quantization footprints (2/2)

- Consider the same signal, now quantized with step **3**. Its histogram is:



- The histogram of the signal quantized with step **2** followed by **3**:



When $a > b$, some bins contain more samples than neighbouring bins. This happens because even bins receive samples from more original bins with respect to the odd bins

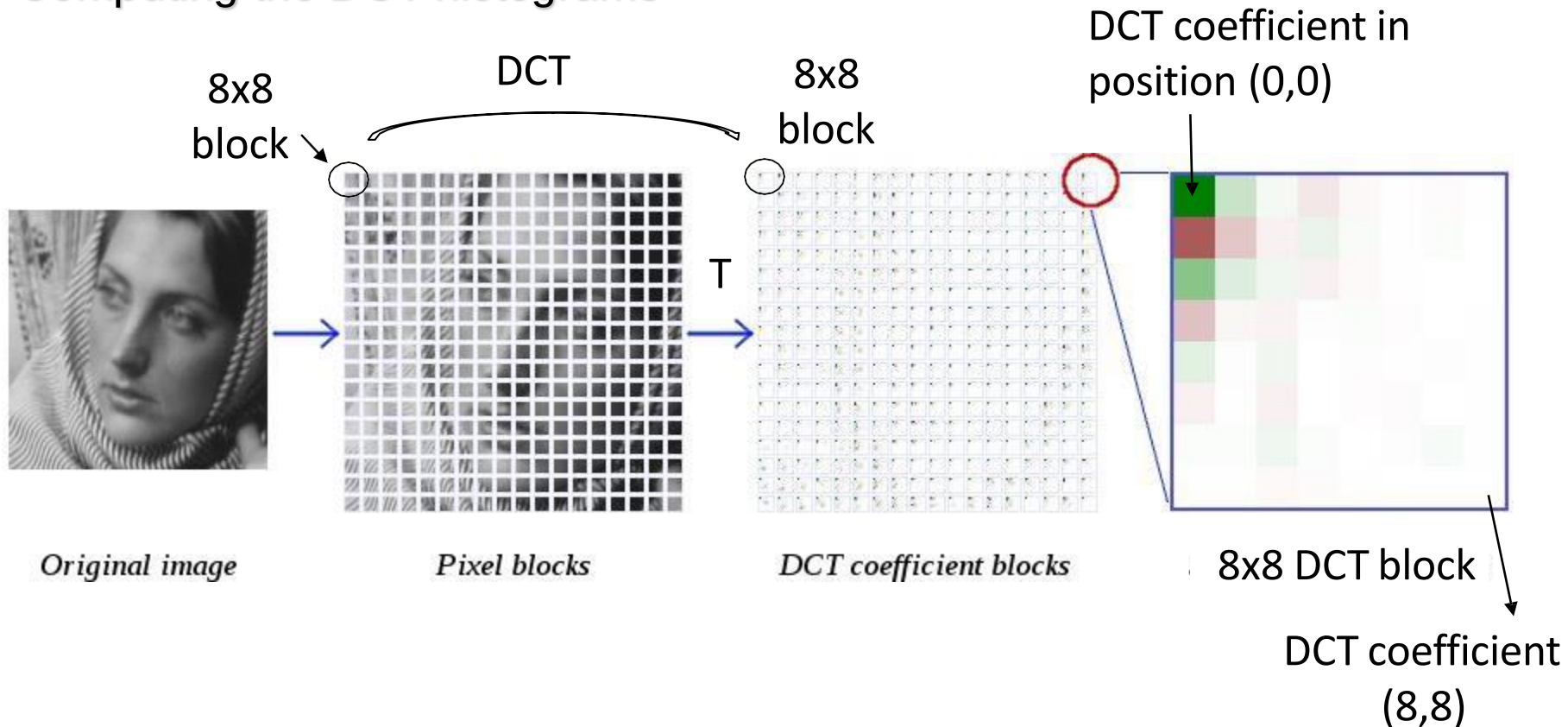
Double JPEG image forensics

Double quantization and DJPEG

- In a JPEG image, quantization is performed in the DCT domain
- Then, in a double-JPEG (D-JPEG) image, the double quantization footprints consist in **periodic artifacts in the histograms of the 8x8 block-DCT coefficients**
 - when $QF_1 < QF_2$, the histograms have periodic holes

Double JPEG image forensics

Computing the DCT histograms



- For each of the 64 DCT coefficients, the histogram of the values taken in all the blocks is computed

Double JPEG image forensics

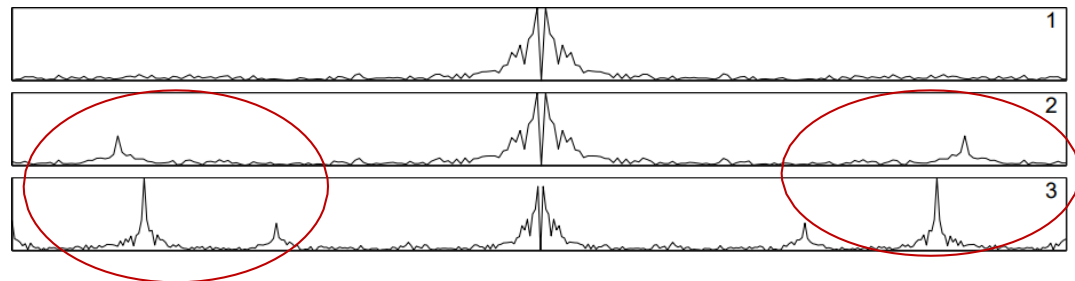
Detection of double quantization

- The periodic patterns are particularly visible in the Fourier domain as strong peaks in the mid and high frequencies
- Then, the Fourier transform of each DCT histogram is evaluated to see if it has certain artifacts [*].
- If the answer is “yes” for at least 1 of the first 10 DCT histograms of the JPEG image, the image is regarded as doubly compressed.
- **Example:** Fourier transform of DCT coeff (1,1)

Single JPEG

Double JPEG (QF1 < QF2)

Double JPEG (QF1 > QF2)



[*] Popescu, Alin C., and Hany Farid. "Statistical tools for digital forensics." *Information Hiding*. Springer Berlin Heidelberg, 2004.

Double JPEG image forensics

Detection of double quantization

- **For the case $QF_1 < QF_2$, the detection is more reliable**

- Peaks and gap are easy to detect

- *Rule of thumb:*

$$\Delta QF = QF_2 - QF_1 \geq 10$$

(the strength of the artifacts depends on Δ)

- $QF_1 < QF_2$ is often the most frequent case in practice

[*] Popescu, Alin C., and Hany Farid. "Statistical tools for digital forensics." *Information Hiding*. Springer Berlin Heidelberg, 2004.

Double JPEG image forensics

Detection of double JPEG compression

■ Several detectors of double JPEG compression

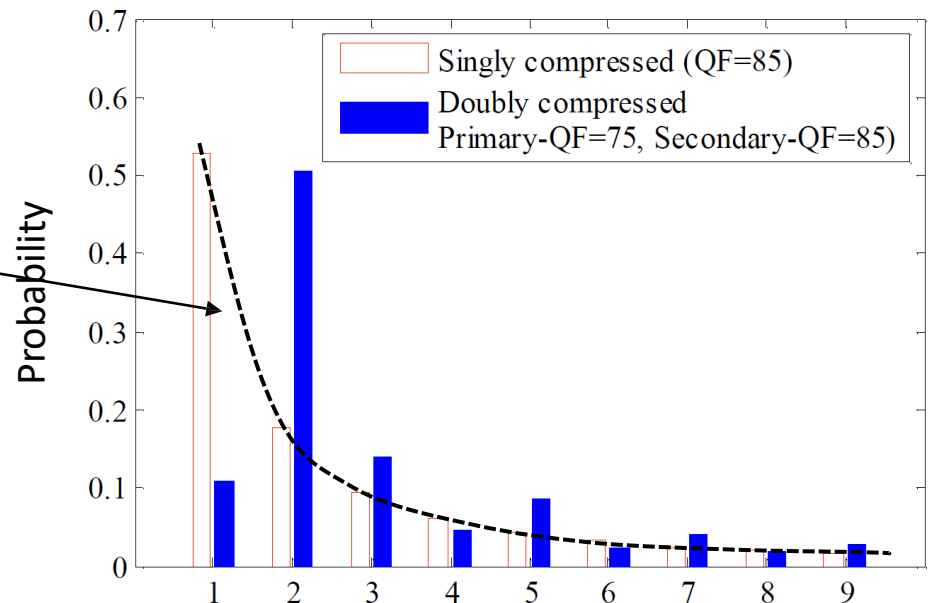
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Double JPEG image forensics

Another feature: FSD distribution

- Another method is based on the **distribution of the First Significant Digits (FSD)** of the block-DCT coefficients.
- For single JPEG images, the distribution of the FSDs follows a known law (*Benford's law*) [**]
- Double compression cause violation of this law
- **Example:**

$$p_i = \log_{10} \left(1 + \frac{1}{i} \right)$$



[**] Fu, Dongdong, Yun Q. Shi, and Wei Su. "A generalized Benford's law for JPEG coefficients and its applications in image forensics." *Electronic Imaging* 2007. International Society for Optics and Photonics, 2007.

Double JPEG image forensics

Beyond model-based approaches

- We have seen examples of *model-based* approaches (relying on statistical models)
- Another category of methods: *data-driven (DD) (or machine learning (ML))* approaches

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Why machine learning?

- **Probabilistic models are often unknown** (in real application scenarios)
- A statistical description may even not be possible
- Then, model-based approaches for data analysis are not viable (possible only under particular conditions)
- **Machine Learning (ML) is about learning structure from data, namely 'examples'**
Example: in a binary classification problem: the statistical formulation of a given phenomenon under H_0 and H_1 is unknown, but samples from the two classes are available

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An example (binary classification)

- Suppose we have 50 photographs/images of elephants (H0) and 50 photos of tigers (H1).



VS



- Now, given a new (different) photograph/image we want to answer the question: is it an elephant or a tiger? [assuming that it is either one or the other]



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Formal formulation of DD/ML setup

- We want the system *to learn* the mapping: $X \rightarrow Y$, where $\mathbf{x} \in X$ is some *object (feature vector)* and $y \in Y$ is a *class label*
- Simplest case: 2-class classification: $\mathbf{x} \in R^n$, $y \in \{\pm 1\}$
- **Training set** (made of labeled examples): $(x^1, y^1), \dots, (x^m, y^m)$
- Generalization purpose: given a previously unseen $\mathbf{x} \in X$, determine $y \in Y$
- **ML methods learn a classification function $y = f(\mathbf{x}, \alpha)$, for a given f , where α is a set of unknown parameters of the function to be optimized**
- These unknown parameters are optimized (“learned”) based on the training set

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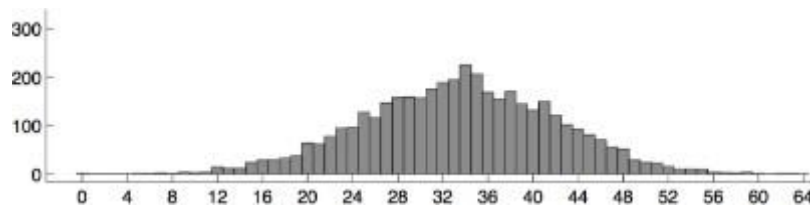
ML algorithms

- **Support Vector Machines (SVM) or Networks**
 - the simplest ML algorithm (one of the most commonly used) for classification and estimation problems
- **Neural Networks (NN)**
 - these networks are usually fed with *feature vectors* ($x \in R^n$ is a feature vector)
- **Deep Neural Networks (DNN), and Convolutional Neural Network (CNN)**
 - Outstanding performance
 - $x \in R^n$ can be an image (image block). The features are *self-learned* by the CNN.

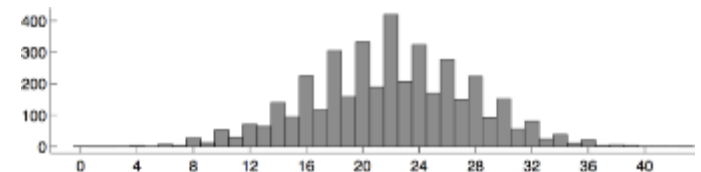
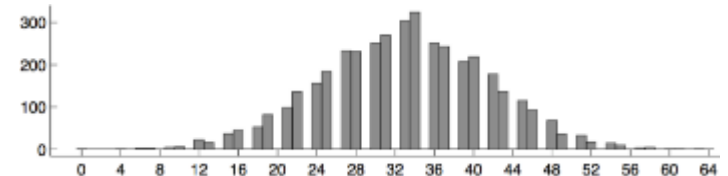
Double JPEG image forensics

SVM-based double JPEG detection

- We can use machine learning techniques to build a classifier that can distinguish between *single JPEG* images (H0) and *double JPEG* images (H1)
- Through SVMs, a detector that can distinguish between single quantized DCT histograms (“without artifacts”) and double quantized DCT histograms (with “artifacts”) can be built



H0

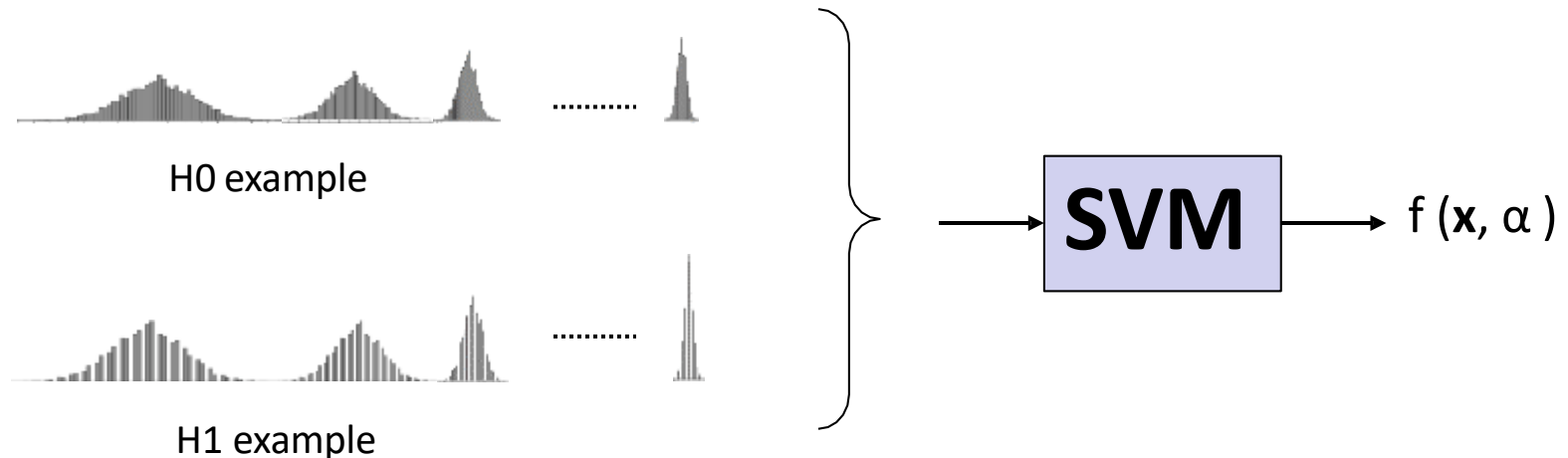


H1

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SVM-based double JPEG detection

- The histograms of the 64 block-DCT coefficients can be concatenated to form a feature vector [***]
- This feature vector can be given as input to an SVM classifier
- **Example** (of input feature vector $\mathbf{x}$):



[***] Pevný, Tomáš, and Jessica Fridrich. "Detection of double-compression in JPEG images for applications in steganography, IEEE Transactions on . Information Forensics and Security vol. 3, No.2 (2008): 247-258

Double JPEG image forensics

Rich feature sets

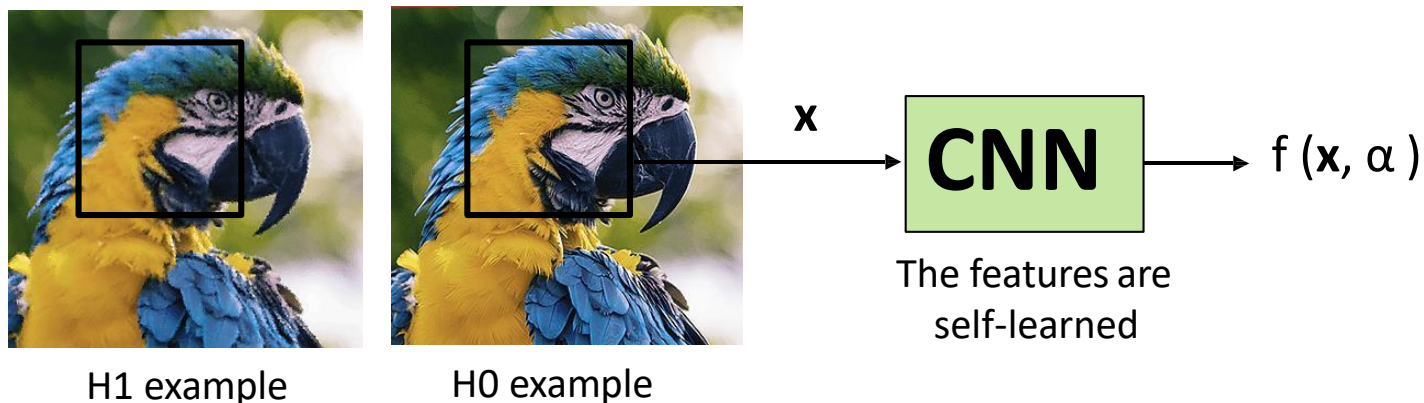
- General **rich sets of features** have been derived in [#], computed from either *the DCT image* and *the pixel image* (first and higher-order features)
- This rich sets of features can be used to train SVM (or NN) models to address several classification task (not only D-JPEG)
 - traces can be captured either by the frequency (DCT) domain features or the pixel domain features
- For D-JPEG detection, even better performance can be obtained (especially, in the most difficult cases, e.g., QF1 $\approx$ QF2)

[#] Jessica Fridrich and Jan Kodovský. "Rich Models for Steganalysis of Digital Images." IEEE Transactions on Information Forensics and Security, 7(3), 868-882, 2012

Double JPEG image forensics

CNN-based D-JPEG detection

- With the adoption of CNN models, it is possible to boost the performance of D-JPEG detection [&]
- A CNN model can be successfully trained, *directly* from the image (or image regions)
- A large amount of training data is needed (representative for all the cases of (QF1,QF2))



[&] Barni, M., Bondi, L., Bonettini, N., Bestagini, P., Costanzo, A., Maggini, M., Tondi, B., Tubaro, S. (2017). Aligned and non-aligned double JPEG detection using convolutional neural networks. *Journal of Visual Communication and Image Representation*, 49, 153-163.

Double JPEG image forensics

Data-driven vs Model-based approaches

Strengths of D-D methods:

- *Much better performance in general*
- *Capable to work under very general conditions.* For Double JPEG detection, a D-D method could work for:
 - $QF1 \{> / <\} QF2$
 - Aligned or not aligned JPEG (the artifacts are different in the aligned and misaligned case)
- *Capable to work in difficult cases ($QF1 \approx QF2$, that is, ΔQF is small)*

Weakness of D-D methods:

- *Are the “learned” features are really peculiar for the detection task under consideration?*
 - D-D solution may rely on *confounding factor*
- *Huge amount of data required (**big-data problem**)*
- *The performance decrease on different image datasets (**dataset mismatch problem**)*
 - Sensitiveness to image properties (e.g., resolution,..)
- *The training phase is very critical*

References

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